Team Purpose and Mission:

- **What questions does your team hope to answer?** We intend to analyze the change in California Fair Plan coverage over the last five years and understand how it correlates to household income and trends in wildfires.
- **What types of data are supposed to be involved?** We will collect data with respect to wildfire frequencies, temporal and geographic distribution of wildfire disasters in California for the past five years. For socio-economic analysis, we are going to summarize the median income and California Fair Plan coverage through zip code & county boundaries.
- **Who is the intended audience?** Stakeholders interested in California Fair Plan coverage and how it subsidizes Californians can use our work product
- **How will your combined skills and interests support the work?** We will be working in both R and Python to pursue geospatial and time series analysis.
- **What can your audience expect in terms of analytical quality and clarity?** A processed csv with all key insights from analysis and dashboard for public usage based off of our synthesized data.

Communication Plan and Collaborative Coding:

- **Which channels will you use for communication and files-sharing?** We will use iMessage for routine and urgent communication. Responses will be expected within one business day. File-sharing will occur through DukeBox and Github based on file size. Additionally, we may use the Google Suite to work on documents and slideshows as necessary.
- **When are members available to work and when will you meet regularly?** Regular meetings will occur during ENV 712 (Thursday afternoons).
- **How will you document decisions and next steps so everyone stays informed?** Tasks will be clarified at the end of the class period, with updates shared every Monday to ensure we have sufficient time to check work before assignments are due on Tuesdays. We will assign tasks via a Github Project Kanban board.

Roles and Responsibilities:

- **What interests each member about the project?** The intersection between finance and the environment, as well as economic health is of interest to the team.
- **What strengths do you bring, and what skills do you want to develop?** Some prior geospatial analysis, but we look forward to building on that. We bring experience in R and Python, so we might have a chance to learn more about the other language.
- **How will you divide responsibilities?** Nigel will lead work in R, Matt will lead work in Python, and Maya can work across the two wherever most needed. Lam will focus on data preparation in Python, cleaning and combining the datasets into the processed CSV, and will support Matt on the geospatial side. Geospatial analysis will be primarily

performed in Python, with time series in R. Our goal will be to create a set of dashboards, one geospatial and one time series to share results and findings. We are willing to be flexible across softwares as needed.

Operating Norms and Ground Rules:

- **How will you prepare for and run meetings?** We plan on starting every Thursday shared session to discuss blockers we're having in our work and decide what the next week of work will be. This will entail a review of our kanban project management board in github which we'll be using as our source of truth on our work progress
- **How will you assign tasks and set deadlines?** We'll use every Thursday as our "sprint" review meeting where we'll divvy up the work to be done in the week to come. We'll also have check-ins on Monday to ensure we're collectively prepared for the submission of our progress on Tuesdays. If there's any urgent issues, we'll coordinate a synchronous time to review the issue at hand on Monday afternoon or Tuesday morning to be prepared to meet the deadline.
- **What should teammates expect when someone completes work independently and uploading to team repository?** We expect a message to be sent out in our group chat.
- **What to do if someone is missing a meeting or deadline?** We will connect with the team and ensure the team member is doing well. We are able to be flexible as other commitments arise (e.g., exam in another class), particularly if communicated ahead of time. That said, if it becomes a continuous issue, we will continue to work with the team member and involve Luana and John as needed.
- **How will you review work before submitting it?** Through Monday check-ins, we can assign review tasks as necessary to ensure we are comfortable with the work before submitting on Thursdays.

Values and Behaviors:

- **What does being a good teammate mean to your group?** We place a strong emphasis on sharing the workload to ensure no single team member is doing too much. Additionally, we value proactive offers to provide help and support during busier times in the semester.
- **Agree on expectations for reliability, respectful communication, constructive feedback, and openness to different ideas.** We will communicate via iMessage, Github updates, and class meetings on Thursdays. Clear, open communication will be emphasized to ensure that everyone's opinion is heard. As we continue to refine our research question(s), we will be sure there is space for everyone to share thoughts.
- **How will you create space for all members to contribute and ask for help?** We are all in a data exploration class and do not assume any expertise in the coding language or statistics generally. We are here to support one another and learn together.
- **How will you address uneven participation or concerns about the quality of someone's work?** We will keep discussions within the group at first, asking the group

member if they are concerned about any aspects of their work and what kind of help may be useful. We will use course material and lectures to help ground our work. If the issue persists, we may involve John and Luana.

Resources and Constraints:

- **What resources can support your work, including course materials, dataset documentation, instructors, TAs, and relevant subject-matter sources?** We will use course materials, including previous lectures, lab workbooks, and assignments. We have substantial time in class with John and Luana each week and can utilize them and the TAs as needed. We will prioritize working with John for Python issues. Tori Yanco is willing to provide help to teams working with California data, so we will reach out to her as needed.
- **What constraints should shape your scope—for example, the six-week timeline, missing data, differences in geographic or temporal coverage, technical skills, or other commitments?** We are aware that the timeline is short, and thus, are limiting the scope of our questions. We expect the Fair Plan and Census data to be relatively clean and will set aside additional time to work with the Cal Fire data. In the first week or two, we are going to be flexible with our question, knowing that data issues may arise.
- **Since follow-up with project partners may not be available, how will you resolve unanswered questions, document assumptions, and communicate limitations in the dashboard?** Being upfront about assumptions and limitations is key. Additionally, if we need to change the scope of our project because of data issues, we are prepared to do so.

Decision-Making Guidelines:

- **How will you make decisions about research questions, datasets, analytical methods, and dashboard design?** We will make decisions collaboratively during our Thursday in-class meetings and Monday online check-ins. We will prioritize approaches that can apply to and align with our research objectives, data that are available and reliable, and the six-week timeline. In the course of our project, we will refine our research questions and analytical methods as we learn more about the datasets.
- **How will you ensure that relevant teammates are involved and that choices are supported by evidence?** We will ensure that all team members have opportunities to contribute their ideas, discuss relevant decisions and contribute to the project based on their own skills and thoughts. We will use data exploration, dataset documentation, course materials, and relevant sources to support our choices. Important decisions and next steps will be documented through our GitHub Project board and shared files.
- **What will you do if you cannot reach consensus?** We will discuss the advantages and limitations of each proposed approach or idea and seek a solution that best meets our shared project goals. If we cannot reach consensus, we will prioritize the most feasible approach based on data availability, theoretical validity, analytical feasibility and time constraints. We will consult the instructor or TA if necessary.

- **When should you seek guidance from an instructor or TA?** We will seek guidance when we encounter persistent data quality issues, uncertainty about appropriate analytical methods, unsolvable technical problems that prevent progress, or disagreements that we cannot resolve within the team. We will also ask for assistance when we need to adjust our research questions or project scope may affect our ability to complete the project on time.

Goals and Objectives:

- **What would a successful project look like?** A successful project will provide a clear overview of changes in California FAIR Plan coverage over the past five years and explore how these changes relate to household income and wildfire trends. We aim to produce a processed CSV dataset, a geospatial dashboard, and a time-series dashboard that communicate our main findings clearly and accurately.
- **Identify a small set of concrete goals and how you will assess them.**
  Data preparation: Integrate and clean the relevant datasets, ensuring that key variables, time periods, and geographic units are sufficiently consistent for comparison.
  Research and analysis: Use appropriate geospatial and time-series analyses to identify patterns in FAIR Plan coverage, household income, and wildfire activity.
  Dashboard development: Create functional and accessible dashboards that directly address our research questions and present our findings through clear maps, charts, labels, and summaries.
  Communication of findings: Explain the main patterns identified, acknowledge data limitations and assumptions, and avoid drawing conclusions that are not supported by the analysis.
  We will assess these goals by reviewing data quality, checking analytical outputs against the source data, and evaluating whether the dashboards communicate the findings in an accurate and interpretable way.
- **What checks will you use to evaluate data quality, dashboard functionality, and the clarity of your findings? Distinguish essential deliverables from optional features.**
  We will check datasets for missing or problematic values, duplicate records, inconsistent variable formats, and potential mismatches in geographic boundaries or time periods. We will verify that summary statistics, charts, and maps accurately reflect the processed data. We will also test dashboard navigation, visual labels, and displayed results to ensure that the information is easy to understand.
  Essential deliverables will include the processed CSV dataset, reproducible analytical workflows, and functional geospatial and time-series dashboards presenting our main findings. Additional interactive features, more detailed analyses, and dashboard enhancements will be optional and pursued only if time and data availability permit.

Schedule and Milestones:

- **Week 1 (10/15-10/22):** Prepare team charter, download data, and start reviewing data

- **Week 2 (10/22-10/29):** Explore and clean data and start to build combined datasets for time series and geospatial analysis. Explore packages to help work with data quickly (e.g., US Census packages)
- **Week 3 (10/29-11/5):** Initial creation of views, tables, and charts to show high-level relationships between data
- **Week 4 (11/5-11/12):** More detailed analysis and minimum viable product of dashboard/deliverable
- **Week 5 (11/12-11/19):** Continue building and validating dashboard, prepare for presention